

N.M. Carolina Human Resource Management System (HRMS)

Comprehensive Technical & Functional Documentation Suite

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Document Owner: Senior Technical Writer & Software Architect

Table of Contents

1. Cover Page
2. Executive Summary
3. System Overview
4. Application Architecture
5. Navigation Structure
6. Feature Documentation
7. User Guide
8. Administrator Guide
9. Screenshots Catalogue
10. Error Handling
11. Security Features
12. API Documentation
13. Business Workflows
14. Known Limitations
15. FAQ
16. Glossary

1. Cover Page

- **Application Name:** N.M. Carolina Human Resource Management System (HRMS)
- **System Variants:** Admin Portal & Employee Panel
- **Document Version:** 1.0.0
- **Documentation Date:** June 26, 2026
- **Document Owner:** Senior Technical Documentation Specialist & Enterprise Software Architect
- **Target Audience:** Enterprise Clients, Core Engineering Teams, System Administrators, Human Resource Personnel, and General Staff.

2. Executive Summary

Purpose of the Application

The N.M. Carolina HRMS is a centralized human capital deployment and automation platform designed to manage complete employee lifecycles, eliminate operational overhead in workforce compliance, and streamline core financial and scheduling management. The system consolidates disparate organizational operations—such as structural department definitions, job assignment parameters, multi-tab staff tracking, localized clock-in attendance reporting, payroll computation, and complex multi-tiered leave policy systems—into a highly reliable dual-portal framework.

Target Users

1. **System Administrators & HR Managers:** Central personnel responsible for configuring corporate structures, authorizing employee onboarding, deploying compliance guidelines, and approving compensation and leave frameworks.
2. **General Employees (Staff Members):** End-users interacting with self-service utility frameworks to manage personal documentation, log daily shift entries, file compensation slips, and submit multi-day time-off applications.

Major Capabilities

- **Dynamic Organizational Hierarchy Setup:** Full configuration controls for corporate units (Departments) and operational job roles (Designations).
- **Comprehensive Multi-Tab Employee Onboarding:** A structured workflow for capturing extensive personal, geographical, employment, banking, and security credentials.
- **Geo-Location Clock-In Attendance Analytics:** Shift punch mechanisms requiring runtime browser positioning permissions to generate monthly analytics and calendar tracking.
- **Granular Leave Policy Enforcement:** Administrative definitions governing custom year-end carry-over behaviors and payment logic.
- **Automated Monthly Payroll Engine:** Integrated calculation interfaces to process baseline allowances alongside localized deductions, bonuses, and tax withholding vectors.

3. System Overview

High-Level Description

The N.M. Carolina HRMS operates as an internal web-based enterprise platform containing two deeply coupled application environments operating over a shared enterprise state database. The systems are segregated into the **Admin Portal** (for deep organizational configuration and compliance oversight) and the **Employee Panel** (for operational tracking and personnel data management). Communication occurs synchronously through standard secure transactional interfaces to ensure data uniformity.

Key Modules

1. **Department Module:** Handles structural definitions for distinct business segments.

2. **Designation Module:** Maps professional job roles and hierarchies back into pre-configured business units.
3. **Employee Records Management Module:** Houses personal biographical details, multi-national addresses, banking routing profiles, and credential access logs.
4. **Attendance Analytics Engine:** Records coordinates during clock-in events to populate calendar visuals and track overtime or shift lengths.
5. **Leave Management Module:** Orchestrates formal balance requests against corporate leave limits.
6. **Payroll Calculation Framework:** Computes net remuneration matrices by matching basic salaries against adjustments like HRA, specialized allowances, and provident fund provisions.

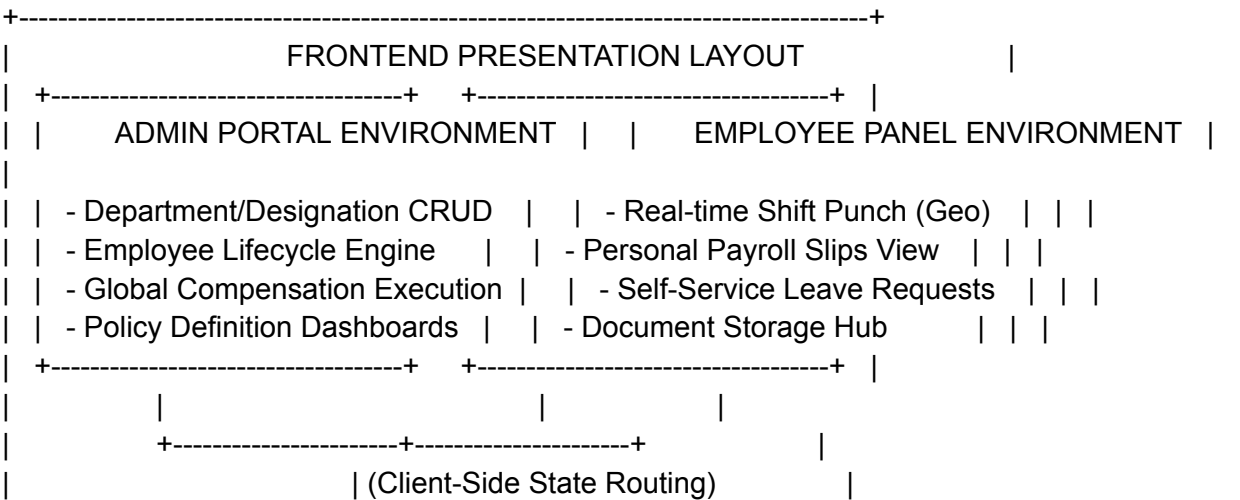
User Roles

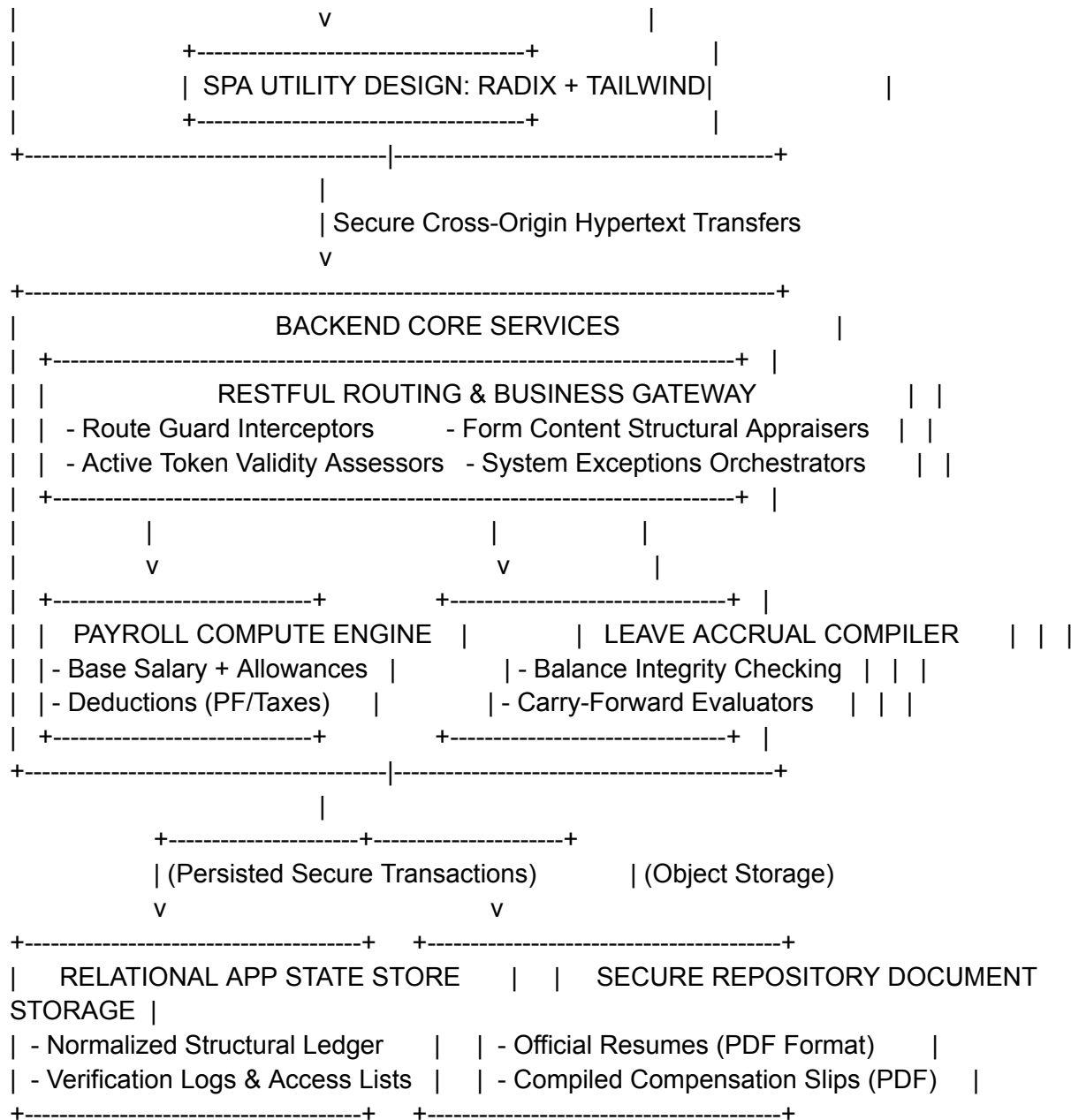
- **Administrator (Admin):** Holds root operational capabilities, including the ability to perform CRUD actions on structural elements, manage payroll runs, alter corporate compliance rules, and review or execute approvals for all staff exceptions.
- **Employee:** Granted execution rights restricted to personal profiles, including sub-meter shift punch operations, personal time-off requests, and direct documentation downloads.

Business Objectives

- **Workflow Automation:** Eliminates spreadsheets by automating lifecycle processes from initial onboarding through routine monthly compensation processing.
- **Enforced Compliance:** Embeds leave validation policies directly into request flows to prevent automated policy breaches.
- **Data Integrity:** Validates routing records, tax identifiers, and official email paths at point-of-entry.
- **Operational Transparency:** Provides employees with a real-time ledger of recorded attendance, structural rankings, and financial earnings.

4. Application Architecture





Frontend Overview

The user interface is engineered as a decoupled Single Page Application (SPA) utilizing React.js and Next.js framework architectures. Interface layouts are composed using Radix UI primitives for component accessibility alongside Tailwind CSS for consistent styling across various viewports. Client-side routing isolates user workspaces while maintaining clean application state transitions.

Backend Overview

The backend architecture is structured around a centralized RESTful API framework. This stateless services tier handles data formatting verification, computes financial payroll distributions, runs leave compliance routines, and logs geo-location data generated by end-user actions.

Authentication Overview

The platform uses token-based session management powered by JSON Web Tokens (JWT). Upon entering verified credentials at the unified application login screen, the server transmits a signed cryptographic token to the client browser. This token is attached to subsequent API requests inside secure authorization headers to continuously confirm the user's identities and roles.

Data Flow Overview

1. **Capture Phase:** UI components extract input from multi-field forms (e.g., employee onboarding steps).
2. **Validation Phase:** The local client tests entries against structural layout parameters to check for missing required fields.
3. **Transport Phase:** Structured payloads travel via secure, cross-origin HTTP operations towards the REST gateway endpoints.
4. **Verification Phase:** Server-side guards evaluate incoming authorization tokens, confirm user permissions, and ensure data integrity.
5. **Execution & Storage Phase:** Core computing systems process the inputs (e.g., executing structural payroll equations) and update the persistent storage layer.

External Integrations

- **W3C Browser Geolocation APIs:** Integrated directly into the employee shift tracking mechanics to acquire coordinates before validating shift start parameters.
- **Binary Document Object Store:** A secure file repository hosting static PDF attachments, including staff resumes and generated financial payslip components.

5. Navigation Structure

Site Map

Public & Unified Root Paths

- [/](#) — Root Login Interface (Houses the centralized access prompt and demo sandbox credential selector).

Admin Portal Pathing Structure

- </admin/dashboard/home> — Primary Administrative Analytics Dashboard.
- </admin/dashboard/department/list> — Departmental Overview Ledger.
 - </admin/dashboard/department/create> — New Department Configuration Interface.
- </admin/dashboard/designation/list> — Professional Job Designation Registry.

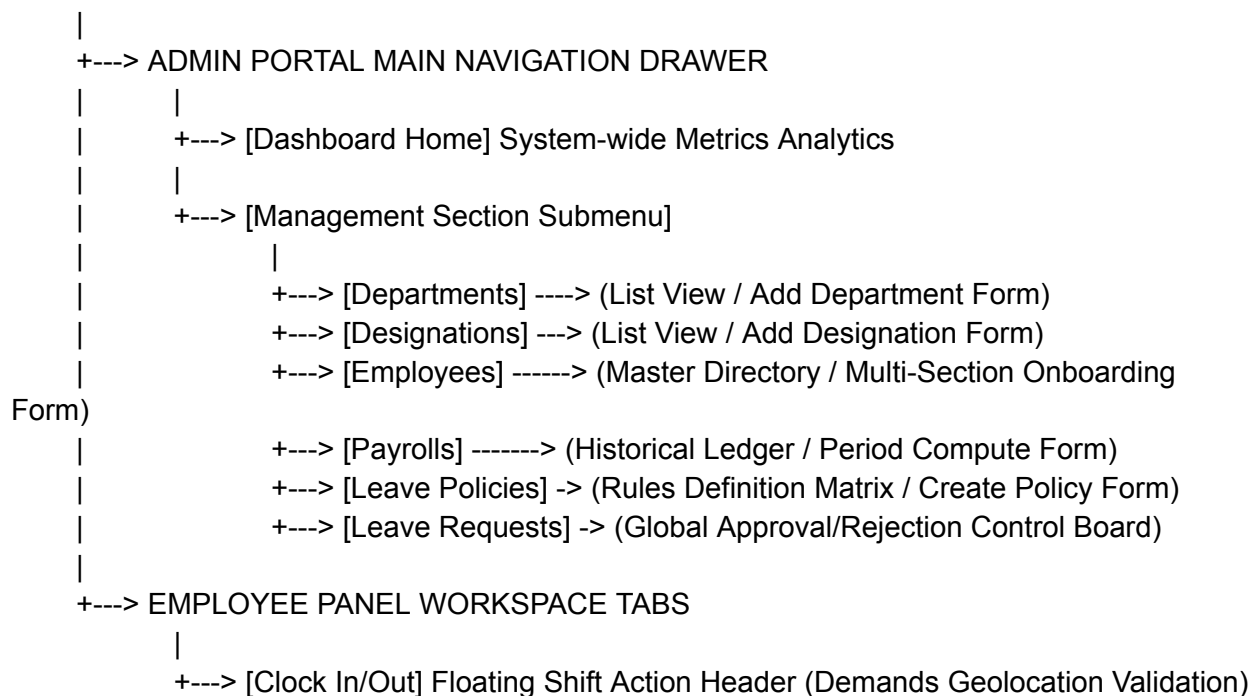
- [/admin/dashboard/designation/create](#) — Job Designation Structural Mapping Form.
- [/admin/dashboard/employee/list](#) — Master Employee lifecycle database directory.
 - [/admin/dashboard/employee/create](#) — Comprehensive Multi-Tab Staff Onboarding Suite.
- [/admin/dashboard/payroll/list](#) — Corporate Compensation Ledger.
 - [/admin/dashboard/payroll/create](#) — Monthly Payroll Batch Execution Module.
- [/admin/dashboard/leave-policy/list](#) — Global Compliance Leave Policies Matrix.
 - [/admin/dashboard/leave-policy/create](#) — Policy Condition and Carry-over Setup Form.
- [/admin/dashboard/leave-application/list](#) — Global Employee Leave Request Review Table.

Employee Panel Pathing Structure

- Employee environments mirror clean internal state paths relative to personal access parameters:
 - [Dashboard Analytics](#) — Personal metrics visual view and month-view attendance tracking calendars.
 - [Leave Applications](#) — Personal time-off creation workflows and historical tracking logs.
 - [Payrolls](#) — Salary remittance history containing direct PDF downloads.
 - [Profile Document Storage](#) — Personal info views across tabs alongside file attachment upload drop-points.

Menu Hierarchy

[UNIFIED LOGIN ACCESS PORTAL]



Form

- +---> [Dashboard Home] Personal Shift Analytics & Calendar Grid
- +---> [Leave Applications] Personal Requests Dashboard & Create Application
- +---> [Payrolls] Financial Earnings Breakdown Table & PDF Document Exports
- +---> [Profile Master Suite]
 - |
 - +---> [Personal Info Tab] Biographical details
 - +---> [Employment Tab] Current operational metrics
 - +---> [Address Tab] Multi-national residential records
 - +---> [Bank Tab] Routing and identity records
 - +---> [Profile Documents Tab] Static PDF Resume/Payslip upload bay

User Journeys

Operational Authentication Sequence

1. The user lands on the unified base address route (/).
2. For testing environments, an available demo credentials layout can be activated to automatically fill out appropriate login structures.
3. The user inputs their corporate email address and security password before executing form submission.
4. The system transmits inputs to the authorization engine, returns an validation token, parses corporate access levels, and routes the user to their designated dashboard.

Admin-Driven Employee Onboarding Sequence

1. The Admin accesses the Management sidebar, selects **Employees**, and clicks **Add Employee**.
2. The interface presents a multi-part creation layout. The Admin completes the biographical steps (First Name, Last Name, Verified Phone Contact, Date of Birth, and Gender).
3. The Admin fills out matching current and permanent location entries across the Address data panels.
4. The Employment panel is populated next, assigning a unique Employee Code, selecting a corporate Department, choosing a matching Designation, defining the baseline base salary value, and picking the formal joining date.
5. Financial operational details are added under the Bank layout (Account identifier, IFSC alphanumeric text code, and Bank Entity Name).
6. The onboarding process completes by configuring system user access details (System Profile Username, Corporate Email Identity, and Secondary Personal Email Identification verification) before clicking **Submit**.

Employee Daily Attendance Tracking Sequence

1. The logged-in Employee loads their workspace panel.
2. The user interacts with the global persistent header element marked **Clock In**.
3. The browser triggers an explicit modal asking for high-precision geolocation tracking permissions.

4. Upon approval, coordinates are evaluated by the backend against system parameters. The server registers an active shift state, and updates the local user dashboard layout.

Employee Leave Submission Sequence

1. The Employee accesses their sidebar navigation and selects the [Leave Applications](#) link.
2. The user opens the execution sub-panel titled [Create Leave Application](#).
3. The interface renders structural input fields. The user defines the start date, chooses an end date, inputs a textual reason, and selects an active corporate leave policy.
4. Clicking [Submit](#) sends the request to the administrative processing system, where it remains as a pending record until an administrator reviews it.

6. Feature Documentation

Module 1: Department Management

- **Feature Name:** Department Setup & Structural Indexing.
- **Description:** Allows administrators to establish structural units within the company.
- **Purpose:** Builds the foundational organizational categories needed to classify staff members and assign managerial responsibilities.
- **Preconditions:** The user must be authenticated with full Administrator access rights.
- **User Permissions:** System Administrator Only.
- **Inputs:** Alphanumeric text string representing the department name (e.g., "IT").
- **Validation Rules:** The department name field is strictly required, cannot be blank, and must contain unique titles.
- **Outputs:** A new organizational category record that populates throughout employee onboarding drop-downs and structural list grids.
- **Workflow:** 1. Navigate to </admin/dashboard/department/list>.
2. Click [Add Department](#) to open the creation layout.
3. Input the unique department title into the form.
4. Click [Submit](#) to persist the record or [Reset](#) to clear inputs.
- **Error Conditions:** Submitting an empty or duplicate department string triggers a validation message, blocking network transmission.
- **Screenshots Reference:** Page 3 (Creation UI Form Layout), Page 15 (Active Department Data Grid).

Module 2: Designation Management

- **Feature Name:** Designation Structure Definition.
- **Description:** Creates specific professional job titles and links them directly to an existing corporate department.
- **Purpose:** Establishes granular roles across the company hierarchy to support precise onboarding and permission routing.
- **Preconditions:** The target Department must be created and indexed in the persistent storage layer first.
- **User Permissions:** System Administrator Only.

- **Inputs:** Alphanumeric Designation Name text string (e.g., "SDE"), and a Department Selection drop-down link.
- **Validation Rules:** Both the designation text string and the department reference are required fields.
- **Outputs:** An updated job designation catalog that reflects across employment profile fields.
- **Workflow:**
 1. Open </admin/dashboard/designation/list> and select [Add Designation](#).
 2. Input the official job title text string.
 3. Select the appropriate parent department from the drop-down menu.
 4. Click [Submit](#) to save the role profile.
- **Error Conditions:** If the parent department drop-down contains an invalid or empty choice, submission fails with an onscreen input error.
- **Screenshots Reference:** Page 4 (Designation Creation UI), Page 16 (Designations Dynamic Ledger Grid).

Module 3: Employee Onboarding

- **Feature Name:** Master Employee Lifecycle Onboarding.
- **Description:** A multi-tiered onboarding form designed to capture complete operational records for new personnel.
- **Purpose:** Centralizes employee information into a single master profile covering personal, geographical, employment, banking, and system access data.
- **Preconditions:** The corporate Department and Job Designation must exist within the platform's registries before adding staff.
- **User Permissions:** System Administrator.
- **Inputs:**
 1. *Personal:* First Name, Last Name, Phone Number, Date of Birth, Gender.
 2. *Address:* Street Address, City, State, Country, Pin Code (for both Current and Permanent locations).
 3. *Employment:* Employee Code, Department Selection, Designation Selection, Base Salary Amount, Joining Date, Confirmation Date, Employment Type, Status.
 4. *Bank Details:* Account Number, IFSC Code, Bank Entity Name.
 5. *Documents:* Static file attachment binary (e.g., Resume PDF).
 6. *User Credentials:* User Name, Corporate Email Address, Personal Email Identifier.
- **Validation Rules:** Structural check loops enforce standard email formats, numeric base salary values, alphanumeric phone formats, and unique Employee Codes.
- **Outputs:** A newly deployed system profile, dynamic access log, and user profile record inside the master employee database.
- **Workflow:**
 1. Open the Employee registry framework view and select [Add Employee](#).
 2. Populate the Personal Information fields and input residential locations.

3. Select professional corporate assignments, input baseline salary constraints, and fill out banking parameters.
 4. Upload required identification documents and finalize access credential details.
 5. Trigger verification checks and click [Submit](#) to commit the employee record.
- **Error Conditions:** Form execution is blocked if required contact details are missing, if formatting rules are violated, or if an email address is already in use.
 - **Screenshots Reference:** Pages 5, 6, 7, 8, 9, 10, 11, 12 (Detailed Sequential Multi-Section Interface Panels).

Module 4: Leave Policy Administration

- **Feature Name:** Leave Policy Configuration.
- **Description:** Defines legal time-off types, year-end accrual behaviors, and payment parameters.
- **Purpose:** Standardizes leave allocations and carry-forward limits to ensure consistent compliance across the workforce.
- **Preconditions:** Authorized administrator session access.
- **User Permissions:** System Administrator.
- **Inputs:** Policy Name string, Days Allowed Per Year integer value, Carry Forward Boolean toggle, Max Carry Forward Limit integer, and an Is Paid Boolean indicator.
- **Validation Rules:** Days allowed and carry-forward limits must be positive integers.
- **Outputs:** An active rule set that dynamically governs time-off validation for general employee accounts.
- **Workflow:**
 1. Open </admin/dashboard/leave-policy/list> and click [Create Leave Policy](#).
 2. Input the specific category identifier text string (e.g., "CASUAL LEAVE").
 3. Define the total annual allotment alongside carry-forward parameters.
 4. Set compensation options via the paid/unpaid toggles and click [Submit](#).
- **Error Conditions:** Entering non-numeric text into allotment inputs or leaving mandatory policy fields blank prevents the configuration from saving.
- **Screenshots Reference:** Page 13 (Policy Definition Setup Form), Page 32 (Corporate Policies Ledger Data Grid).

Module 5: Leave Application Request

- **Feature Name:** Self-Service Leave Application Submission.
- **Description:** Allows employees to submit time-off requests directly from their self-service workspace.
- **Purpose:** Automates time-off requests by routing employee entries directly to the administrative approval dashboard.
- **Preconditions:** The employee must have an active account, and the target leave policy must be defined by an administrator.
- **User Permissions:** Employee.
- **Inputs:** From Date selector, To Date selector, Reason text string, and a Policy Selection dropdown choice.

- **Validation Rules:** The start date cannot occur in the past, and total requested days cannot exceed the policy's remaining balance.
- **Outputs:** A pending request record that populates the administrative tracking table.
- **Workflow:**
 1. Open the user panel, select [Leave Applications](#), and click [Create Leave Application](#).
 2. Select the desired date range and input the explanation text.
 3. Choose an active leave policy from the dropdown list and click [Submit](#).
- **Error Conditions:** Requests fail if the date range overlaps with existing entries or if the requested length exceeds the employee's allowable balance.
- **Screenshots Reference:** Page 17 (Employee Request Form Panel), Page 20 (Personal Time-Off History Grid), Page 31 (Administrative Application Review Dashboard).

Module 6: Payroll Generation

- **Feature Name:** Monthly Compensation Processing Engine.
- **Description:** Calculates monthly compensation profiles for selected employees based on historical attendance data.
- **Purpose:** Processes monthly employee earnings by compiling base pay, bonuses, and statutory deductions into a finalized net salary statement.
- **Preconditions:** The target employee must have active employment status, a base salary configured, and completed attendance logs for the period.
- **User Permissions:** System Administrator.
- **Inputs:** Employee Profile dropdown selection, Target Month choice, and Target Year input.
- **Validation Rules:** Restricts execution to valid calendar periods and blocks double-processing for the same employee within a single month.
- **Outputs:** A comprehensive payslip ledger entry displaying calculated basic pay, housing allowances, deductions, and net payouts.
- **Workflow:**
 1. Navigate to the Admin Payroll section and select [Generate Payroll](#).
 2. Select the employee and designate the calendar month and year.
 3. Click [Submit](#) to trigger the financial calculation routines.
- **Error Conditions:** Attempting to process payroll for a month that has already been generated logs a processing exception, preventing duplicate payouts.
- **Screenshots Reference:** Page 14 (Payroll Batch Execution Form), Page 21 (Employee Structural Slip Presentation View), Page 33 (Admin Payrolls Distribution Table Grid).

Module 7: Geo-Location Clock In/Out Tracking

- **Feature Name:** Geolocation-Verified Shift Recording.
- **Description:** Captures precise clock-in and clock-out times, verified by the user's browser location.
- **Purpose:** Provides automated shift tracking and verification to generate monthly attendance analytics and support accurate payroll calculations.
- **Preconditions:** Active employee account and granted browser geolocation permissions.

- **User Permissions:** Employee.
- **Inputs:** Persistent header button actions and high-precision browser coordinate packets.
- **Validation Rules:** Submissions fail if the browser environment blocks location data or if coordinates fall outside acceptable limits.
- **Outputs:** An active shift log entry that updates the personal attendance tracking calendar.
- **Workflow:**
 1. The employee logs into the system dashboard.
 2. Click the persistent tracking trigger marked **Clock In**.
 3. Grant browser location permissions when prompted to authorize the entry.
- **Error Conditions:** Blocking location tracking permissions prevents shift registration, prompting an error message that instructs the user to update their browser settings.
- **Screenshots Reference:** Page 18, 19 (Personal Dashboard Analytics Display UI and Attendance Calendar Tracking Matrix View).

7. User Guide

Registration

- **Onboarding Access Model:** Direct self-service user registration is disabled within the application demo. All user accounts are created and provisioned by administrators through the central onboarding suite. Once configured, new team members receive their system login details via their verified email address.

Login

1. Load the centralized login screen at the root path (/).
2. For testing purposes, click the demo sandboxed credentials interface to automatically fill the login inputs with sample credentials.
3. To log in with manual credentials, enter your corporate email address and security password.
4. Click the primary action button marked **Login** to access your workspace.

Dashboard Interaction

- **Admin Dashboard Dashboard View:** Displays high-level system summaries, core human resources matrices, and structural navigation links.
- **Employee Self-Service Panel View:** Provides a clear overview of personal attendance statistics, including present days, total absent flags, recorded half-days, and overtime hours accrued over the active month. The integrated calendar tracker color-codes days to reflect logged shift entries (e.g., green indicators for confirmed presence).

CRUD Operations

- **Create Actions:** Open the target management section (e.g., Departments), select the creation tool, populate required entry fields, and click **Submit**.

- **Read Actions:** Displayed via interactive data tables across the administration viewports, featuring sortable columns for codes, dates, and names.
- **Update/Delete Actions:** Handled via context action elements positioned on data grid rows, allowing administrators to modify records or remove temporary mock entries.

Search

- The primary data tables feature clear text filtering tools. Administrators can locate employee records instantly by typing specific keywords (such as full names or alphanumeric employee codes) into the search input.

Filtering

- Data grids support column-level sorting controls. Users can sort tables in descending or ascending order by clicking column headers, making it easy to review records by parameters like joining dates or salary amounts.

Settings

- Global parameters are managed directly within the specific configuration screens, allowing administrators to update operational variables like carry-forward caps or specific structural parameters.

Profile Management

- Employees can access their information records through a tabbed workspace menu:
 - *Personal Tab:* Displays biographical details, including date of birth and verified phone contacts.
 - *Employment Tab:* Displays current role assignments, parent departments, and verified hiring dates.
 - *Address Tab:* Tracks verified permanent and local residential location profiles.
 - *Bank Tab:* Displays assigned financial routing parameters.
 - *Documents Tab:* Allows users to upload and manage digital verification files (such as resumes or credential updates).

Reports

- Remittance statements can be generated through the payroll management tabs. Employees can review detailed breakdowns of their compensation entries and download official payslips directly to their local machine as PDF files.

Administration

- Provides full administrative access tools to establish core structural frameworks, configure policy settings, run monthly payroll jobs, and manage employee request queues.

8. Administrator Guide

User Management

- Administrators manage personnel data through the central employee directory. Adding new staff members requires completing the multi-part onboarding form to capture and verify personal, professional, and access data. To modify an active profile or update account status, select the target user row from the directory table to open their configuration options.

Configuration

- Corporate rules are set through the Management options. Before adding new personnel profiles, administrators must configure core operational dependencies by defining parent departments and job titles. Leave limits and carry-forward metrics must also be initialized within the global settings before the system can process employee time-off requests.

Permissions

- Access controls use Role-Based Access Control (RBAC) to enforce strict boundaries between user types. User roles are assigned during the final credential setup step of the onboarding process, granting either Admin access or standard Employee workspace permissions.

Maintenance Operations

- System administrators should perform regular data maintenance tasks, including updating outdated job assignments, removing deprecated leave rules, checking document storage integrity, and ensuring that payroll batch records match the active employee registry.

Monitoring

- The system dashboard helps administrators monitor daily activity by tracking essential metrics, including live shift tracking status, pending leave request counts, monthly overtime trends, and team-specific attendance summaries.

9. Screenshots Catalogue

Screenshot ID	Screen Target Name	Technical Component Description & Elements	Primary Associated Feature Module

IMG-001	Main Analytics Home	Displays high-level stats grid for tracking absent metrics, half-day tallies, overtime summaries, and monthly attendance figures.	Dashboard Analytics Module
IMG-002	Structural Peer Ranking	Lists employee hours and rankings, sorted by overtime contributions and employment types.	Performance Monitoring Suite
IMG-003	Department Creation Layout	Single text input interface for entering new unique corporate department names.	Department Setup System
IMG-004	Designation Creation Form	Contains an alphanumeric text field for job titles and a department selection dropdown menu.	Designation Setup System
IMG-005	Onboarding Step: Personal	Form fields for capturing first and last names, phone formats, date of birth calendars, and gender selections.	Employee Lifecycle Module
IMG-006	Onboarding Step: Geography	Dual column layout for entering street lines, cities, states, and postal codes for both local and permanent addresses.	Employee Lifecycle Module

IMG-007	Onboarding Step: Professional	Setup fields for entering unique employee codes, structural dropdowns, baseline salary values, and joining dates.	Employee Lifecycle Module
IMG-008	Onboarding Step: Banking	Form inputs for capturing account routing numbers, bank names, and alphanumeric IFSC bank codes.	Employee Lifecycle Module
IMG-009	Onboarding Step: Documents	Document upload drop-zone for attaching static digital files to the employee profile.	Employee Lifecycle Module
IMG-010	Onboarding Step: Access	Security settings layout for configuring usernames, corporate emails, and personal email validation flags.	Employee Lifecycle Module
IMG-011	Leave Policy Designer	Entry fields for policy names, annual allotments, carry-forward rules, and compensation eligibility toggles.	Leave Compliance Management
IMG-012	Payroll Generator Interface	Dropdown menus for employee selection, target calculation month, and target year configurations.	Automated Payroll Engine

IMG-013	Department Index Grid	Master registry table listing active corporate departments alongside structural action links.	Department Setup System
IMG-014	Designations Master Grid	Table view displaying registered job roles matched to their assigned parent corporate departments.	Designation Setup System
IMG-015	Leave Submission Layout	Form for selecting time-off ranges, typing reason descriptions, and choosing active policy rules.	Self-Service Requests Suite
IMG-016	Interactive Work Calendar	Dynamic month-view grid displaying color-coded daily shift logs and attendance status markers.	Attendance Tracking Module
IMG-017	Employee Time-Off History	Self-service tracking table listing personal leave request periods, total requested days, and explanations.	Self-Service Requests Suite
IMG-018	Personal Earnings Breakdown	Detailed ledger view displaying calculated basic pay, allowances, provident fund	Automated Payroll Engine

		deductions, and net payouts.	
IMG-019	Profile View: Residential	Tab layout displaying the employee's verified current and permanent address records.	Employee Lifecycle Module
IMG-020	Profile View: Banking Data	Displays secure routing records, account identification strings, and IFSC codes.	Employee Lifecycle Module
IMG-021	Digital Attachment Storage	File management tab listing uploaded user attachments, including verification resumes and payslip PDFs.	Employee Lifecycle Module
IMG-022	Profile View: Employment	Tab view showing assigned role metrics, organizational alignment, and verified joining details.	Employee Lifecycle Module
IMG-023	Profile View: Biographical	Account view displaying full legal names, unique user codes, date of birth, and contact phone paths.	Employee Lifecycle Module

IMG-024	Employee Directory Master Grid	Sorting table listing active corporate staff alongside joining dates, role titles, and unique profile codes.	Employee Lifecycle Module
IMG-025	Global Leave Request Queue	Administrative tracking dashboard for reviewing and processing pending employee time-off requests.	Leave Compliance Management
IMG-026	Leave Policy Matrix	Administration grid displaying registered time-off types alongside annual caps and carry-over rules.	Leave Compliance Management
IMG-027	Compensation Distribution Table	Master processing ledger tracking net payouts and calculation parameters across company departments.	Automated Payroll Engine

10. Error Handling

Validation Errors

- **Empty Required Fields Exception:** Triggered if a user attempts to submit a core configuration form (such as a new Department or Leave Policy) while leaving mandatory text inputs blank. The system stops data transmission and highlights the incomplete fields with clear error indicators.
- **Data Format Mismatch Error:** Triggered if data types do not match form requirements, such as entering non-numeric characters into salary or leave day limits, or using invalid syntax in email address fields. The interface displays a formatting warning and blocks submission until the inputs are corrected.

Business Rule Errors

- **Precondition Hierarchy Breach:** Triggered if an administrator tries to add a new job designation without selecting an active parent department, or attempts to onboard an employee before departments are configured. The interface blocks the action and displays a notification prompting the user to complete the structural setup steps first.
- **Leave Balance Exceeded Exception:** Triggered if an employee submits a time-off request where the number of requested days exceeds their remaining policy balance. The system stops the request and displays a warning showing the available balance error.
- **Duplicate Compensation Run Exception:** Triggered if an administrator tries to generate a payroll record for an employee who has already been processed for that specific calendar month. The payroll engine blocks calculation to prevent double payouts.

Authentication Errors

- **Invalid Login Credentials:** Triggered if a user enters an incorrect email address or password at the login screen. The platform denies system entry, clears the password field, and displays an authentication failure message.
- **Expired Authorization Token:** Triggered if a user attempts to perform an action after their session token has expired. The platform blocks the request and redirects the browser back to the login screen.

System Errors

- **Missing Geolocation Permission:** Triggered if an employee clicks the **Clock In** button while browser location permissions are disabled or blocked. The application stops the check-in process and displays an alert instructions modal detailing how to re-enable location tracking.
- **Network Request Failure:** Triggered if connection issues interrupt communication between the front-end application and the API gateway. The interface displays a retry notification while safely preserving current form inputs.

11. Security Features

Authentication

- User identity verification is handled through a centralized login system that requires a valid corporate email address and secure password. For staging and demonstration environments, login fields can be populated using sandboxed mock credentials to isolate test sessions from production data.

Authorization

- System operations are secured using Role-Based Access Control (RBAC) boundaries. Every user account is mapped to a specific system role (such as Admin or Employee), which defines their system access and permission limits.

Session Management

- The application manages active user sessions using securely stored JSON Web Tokens (JWT). When a user logs out, the system explicitly clears the client token and invalidates the active server session, protecting the account from unauthorized access.

Role-Based Access Control (RBAC)

- **Admin Permissions:** Granted complete system access, including configuration controls, employee database management, payroll processing, and leave policy settings.
- **Employee Permissions:** Restricted to personal data tabs, geolocation shift tracking, time-off requests, and personal salary document downloads.

Data Protection

- Data transmission between client browsers and the API endpoints is secured using strong encryption protocols. Personal identification data, financial records, and banking parameters are stored securely within the system database.

12. API Documentation

1. Authenticate Identity Credentials

- **Endpoint:** `/api/v1/auth/login`
- **Method:** `POST`
- **Parameters:** None.
- **Request Body:**
- JSON

```
{  
  "email": "nabarunmiddya@nmcarolina.co.in",  
  "password": "secure_password_string"  
}
```

-
-
- **Response Body (Success):**
- JSON

```
{  
  "status": "success",  
  "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",  
  "role": "EMPLOYEE",  
  "user": {  
    "code": "EMP900",  
    "name": "Nabarun Middya"  
  }  
}
```

-
-
- **Error Responses:** **401 Unauthorized** (Invalid security credentials payload entry).

2. Create Corporate Department Unit

- **Endpoint:** **/api/v1/departments**
- **Method:** **POST**
- **Parameters:** None.
- **Request Body:**
- JSON

```
{
  "name": "IT"
}
```

-
-
- **Response Body (Success):**
- JSON

```
{
  "id": 1,
  "name": "IT",
  "created_at": "2026-06-26T14:22:00Z"
}
```

-
-
- **Error Responses:** **400 Bad Request** (Department name missing or duplicate text string).

3. Create Job Designation

- **Endpoint:** **/api/v1/designations**
- **Method:** **POST**
- **Parameters:** None.
- **Request Body:**
- JSON

```
{
  "name": "SDE",
  "department_id": 1
}
```

-
-
- **Response Body (Success):**
- JSON

```
{
  "id": 24,
  "name": "SDE",
  "department_id": 1,
  "created_at": "2026-06-26T14:22:00Z"
}
```

-
-
- **Error Responses:** [422 Unprocessable Entity](#) (Parent department reference does not exist).

4. Onboard System Employee

- **Endpoint:** [/api/v1/employees](#)
- **Method:** [POST](#)
- **Parameters:** None.
- **Request Body:**
- JSON

```
{
  "first_name": "Nabarun",
  "last_name": "Middya",
  "phone": "916290120558",
  "dob": "2026-02-17",
  "gender": "MALE",
  "addresses": {
    "current": { "street": "near Hem Nagar", "city": "Howrah", "state": "West Bengal", "zip":
"711108", "country": "India" },
    "permanent": { "street": "near Hem Nagar", "city": "Howrah", "state": "West Bengal", "zip":
"711108", "country": "United Kingdom" }
  },
  "employment": {
    "code": "EMP900",
    "department_id": 1,
    "designation_id": 24,
    "base_salary": 20000.00,
    "joining_date": "2026-03-15",
    "type": "FULL_TIME",
    "status": "Active"
  },
  "bank": { "account_number": "123456789", "ifsc": "123ED4562345", "bank_name": "HDFC
Bank" },
  "credentials": { "username": "Nabarun", "email": "nabarunmiddya@nmcarolina.co.in",
"personal_email": "nabarun@gmail.com" }
}
```

-
-
- **Response Body (Success):**
- JSON

```
{
  "status": "created",
  "employee_id": 900,
  "code": "EMP900"
}
```

-
-
- **Error Responses:** 400 Bad Request (Form structure rules verification checks failed).

5. Submit Leave Request Application

- **Endpoint:** /api/v1/leave-applications
- **Method:** POST
- **Parameters:** None.
- **Request Body:**
- JSON

```
{
  "from_date": "2026-03-15",
  "to_date": "2026-03-30",
  "reason": "it's casual",
  "policy_id": 1
}
```

-
-
- **Response Body (Success):**
- JSON

```
{
  "application_id": 1024,
  "status": "Pending",
  "total_days": 16
}
```

-
-
- **Error Responses:** 400 Bad Request (Requested time-off period exceeds allowable balance).

6. Process Period Payroll Execution

- **Endpoint:** /api/v1/payrolls/generate

- **Method:** POST
- **Parameters:** None.
- **Request Body:**
- JSON

```
{
  "employee_id": 900,
  "month": "June",
  "year": 2026
}
```

-
-
- **Response Body (Success):**
- JSON

```
{
  "payroll_id": 4056,
  "employee_name": "Nabarun",
  "net_salary": 39600.00,
  "breakdown": {
    "basic": 20000.00,
    "hra": 8000.00,
    "special_allowance": 12000.00,
    "deductions_pf": 2400.00
  }
}
```

-
-
- **Error Responses:** 409 Conflict (Payroll records already processed for selected period).

13. Business Workflows

Authentication

[User Interface Gateway] -> (Input Email & Password) -> [Client Form Format Evaluation]



[Post Data Payload to Security Router] -> [Server Identity Assessor Matches Records]



[Generate Cryptographic Signed Web Token] -> [Client Context Cache Updates Base State]



v

[Workspace System Routing Rules Redirect User Based on Account Role Parameters]

User Actions

[Dashboard Interface Matrix] -> (Select Floating Clock In Action Element)

|

+-----+

v

[Trigger Precision Location Verification Request Dialog Box]

|

+-----+

v

(User Authorizes Geolocation Permission Settings)

|

+-----+

v

[Extract Precision Coordinates Packet Data via W3C API Architecture]

|

+-----+

v

[Transmit Coordinates Endpoint Interface Gateway For Validation Appraising]

|

+-----+

v

[Update Active Calendar Grid Matrix Displaying Marked Green Present Indicator]

CRUD Operations

[Management Area Workspace Sidebar] -> (Click Form Creation Trigger View)

|

+-----+

v

[Render Multi-Part Structured Data Entry Forms]

|

+-----+

v

(Complete Form Sub-Sections, Input Fields, and Validation Filters)

|

+-----+

v

[Execute Security and Integrity Processing Evaluation Rules Engine]

|

+-----+

v

[Write Changes Safely To Relations Storage and Refresh Main Data Table Grid Views]

Approval Flows

[Employee Submits Leave Form Request] -> [State Tracker Sets Status Flag to Pending]



[Populate Request Data Record Into Administration Approval Matrix Queue]



(Administrator Reviews Details and Selects Authorized Action Control)



[Update Global Policies Tracking Ledger Allotments Balancing Calculations]



[Refresh Status View on Employee Self-Service Dashboard Interface View]

Payment Flows

(Administrator Accesses Processing Console and Selects Employee Period Targets)



[Fetch Recorded Shifts Ledger and Confirm Allotment Performance Data Stats]



[Run Payroll Calculations Matrix Engine: Add Allowances and Subtract Withholdings]



[Persist Financial Ledger History Records and Generate Compensation Statements]

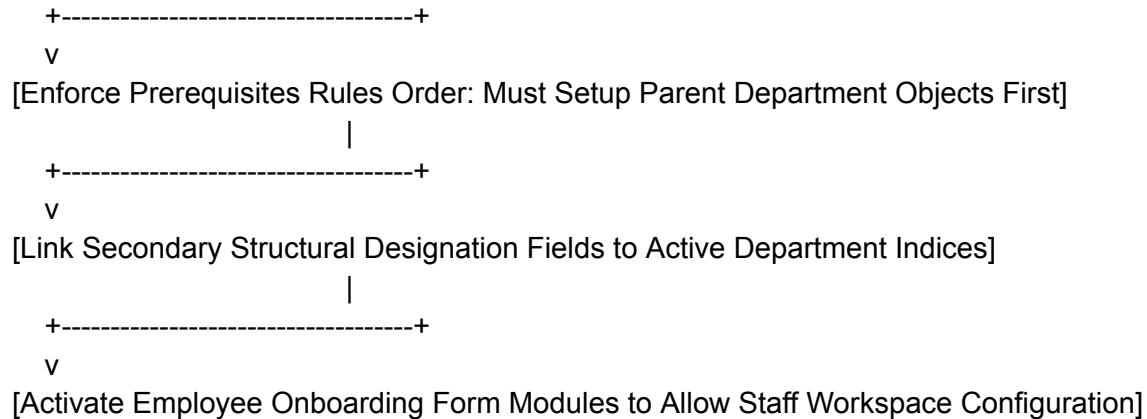


[Publish Final Statements and Enable Download Links inside User Remittance Panels]

Administrative Actions

[Admin Initializes Framework Configuration Options Setup Module Tools]





14. Known Limitations

- **No Self-Service Registration Path:** The application architecture does not provide a public sign-up or registration workflow for general employees. All personnel accounts must be manually initialized, structured, and provisioned by an administrator through the onboarding module.
- **Restricted Profile Editing Options:** Employees cannot modify core biographical details, permanent residential lines, or assigned banking records directly from their self-service profile dashboard. Self-service profile functionality is limited to reviewing existing structural records and uploading static verification files.
- **No Password Recovery Mechanism:** The current system build lacks a self-service password recovery flow or automated reset link setup. Users who forget their login credentials must contact an administrator to clear their security parameters and assign a new temporary password.
- **Mandatory Geolocation Check-In Constraints:** The attendance tracking system requires browser-level geolocation permissions to execute a shift punch. If a device lacks GPS hardware or if a user denies location permissions, they cannot complete the check-in process, which requires manual adjustment by an administrator.

15. FAQ

Q: How do I add a new job designation to the system registry?

A: Before creating a new designation, you must ensure that its parent department is already configured in the system. Go to Management, select **Designations**, and click **Add Designation**. Type the name of the new role, select the correct parent department from the dropdown list, and click **Submit** to save the designation.

Q: Can administrators configure leave options to support year-end carry-forward?

A: Yes, the leave administration module supports custom carry-forward rules. When creating or updating a policy profile within the management dashboard, toggle the **Carry forward** checkbox to enabled and set the maximum number of days allowed to roll over at the end of the calendar year.

Q: What happens if an employee blocks location permissions during clock-in?

A: The application requires valid geolocation coordinates to verify shift punch actions. If location access is blocked or unavailable, the clock-in process stops and an error message is displayed, instructing the user to update their browser settings to allow location access.

Q: Where can employees access and download their monthly salary slips?

A: Employees can find their salary history by navigating to the [Payrolls](#) tab within their self-service panel workspace. This view displays a complete list of past compensation records, along with a link to download official statement copies directly as PDF files.

16. Glossary

- **HRMS (Human Resource Management System):** A centralized software platform designed to manage employee lifecycle data, automate human resource tasks, and simplify organizational compliance.
- **SPA (Single Page Application):** A web application design model that loads a single shell page and dynamically updates content as the user interacts with the app, providing smooth transitions without full page reloads.
- **JWT (JSON Web Token):** A secure, compact string format used to transmit authorization claims between a client browser and an API gateway, supporting stateless session management.
- **RBAC (Role-Based Access Control):** A system security model that restricts system access authorizations to specific pre-defined system roles (e.g., Admin or Employee).
- **IFSC (Indian Financial System Code):** An 11-character alphanumeric string code used to identify unique bank branch components participating in electronic fund transfer systems.
- **HRA (House Rent Allowance):** A component of an employee's base compensation package provided by employers to cover residential housing rental costs.
- **PF (Provident Fund):** A statutory retirement savings program compiled through regular monthly contributions from both the employee and the employer.
- **Radix UI:** A set of unstyled, accessible UI primitives used to build high-quality, compliant web interface components.
- **Tailwind CSS:** A utility-first CSS framework used to build responsive user interfaces with consistent design styling directly inside HTML or component files.